Received money from ₹10 = 10 × 20 = ₹200
Remaining money = 1840 – 200 = ₹1640
If remaining 30 note of ₹50, then received money = 30 × 50 = ₹1500.
This condition is not possible.
Hence, statement I, II and III are not true.

9. Which number amongst 2^{40}, 3^{21}, 4^{18} and 8^{12} is the smallest?

(a) 2^{40} (b) 3^{21}
(c) 4^{18} (d) 8^{12}

Ans. (b) : According to the question,

$4^{18} = 2^{36}$

$8^{12} = 2^{36}$

Let $x = 2^{40}$

$\log_{10}x = 40 \log_{10}2$

$= 40 \times 0.3010 = 12.04$

Let $y = 3^{21}$

$\log_{10}y = 21 \log_{10}3$

$= 21 \times 0.477 = 10.017$

So, the smallest number $= 3^{21}$

10. The digit 1 to 9 are arranged in three rows in such a way that each row contains three digits, and the number formed in the second for in twice the number formed in the first row; and the number formed in the third row in thrice number formed in the first row. Repetitions of digits is not allowed. If only three of the four digits 2, 3, 7 and 9 are allowed to used in the first row, how many such combination are possible to be arranged in the three rows ?

(a) 4 (b) 3
(c) 2 (d) 1

Ans. (c) : Numbers arranged in the first row cannot be started with 9 or 7 because if they are doubled or tripled, they will become four digits.
Unique digit is to be taken each time while forming the number, there should be no repetition.
So we can take 273 and 327 in the first row.
So, total 2 combinations are possible.

Directions for the following 3 (three) items:

Read the following two passages and answer the items that follow the passages. Your answers tothese items should be based on the passages only.

Passage -1

"In simple matters like shoe-making, we think only a specially trained person will serve our purpose, but in politics, we presume that everyone who knows how to get votes knows how to administer a State. When we are ill, we call for a trained physician, whose degree is a guarantee of specific preparation and technical competence-we do not ask for the handsomest physician, or the most eloquent one : well then, when the whole state is ill should we not look for the service and guidance of the wisest and the best?"

11. Which one of the following statements best reflects the message of the author of the passage ?

(a) We assume that in a democracy any politician is qualified to administer a State
(b) Politicians should be selected from those trained in administration
(c) We need to devise a method of barring incompetence from public office
(d) As voters select their administrators, the eligibility of politicians to administer a State cannot be questioned

Ans. (b) : Through the various examples and the question asked by the author at the end reveals the authors message, which conveys the message that politicians should be chosen from amongst those trained in administration.

Passage -2

The poverty line is quite unsatisfactory when it comes to grasping the extent of poverty in Indian. It is not only because of its extremely narrow definition of 'who is poor' and the debatable methodology used to count the poor, but also because of a more fundamental assumption underlying it. It exclusively relies in the notion of poverty as insufficient income or insufficient purchasing power. One can better categorize it by calling it income poverty. It poverty is ultimately about deprivations affecting human well-being, then income poverty is only one aspect of it. Poverty of a life, in our view, lies not merely in the impoverished state in which the person actually lives, but also in the lack of real opportunity given by social constraints as well as personal circumstances- to choose other types of living. Even the relevance of low incomes, meagre possessions, and other aspects of what are standardly seen as economic poverty relate ultimately to their role in curtailing capabilities, i.e. their role in severely restricting the choices people have to lead variable and valued lives.

12. Why is the methodology adopted in India to count the 'poor' debatable?

(a) There is some confusion regarding what should constitute the poverty line'
(b) There are wide diversities in the condition of the rural and urban poor

(c) There is no uniform global standard form measuring income poverty
(d) It is based on the proposition of poverty as meagre income of buying capacity

Ans. (d) : It is clear from the passage that the reason for the controversial methodology of calculation of 'poor' adopted in India is based on presenting poverty as insufficient income or insufficient purchasing power.

13. Why is income poverty only one measure of counting the 'poor'?

(a) It talks of only one kind of deprivation ignoring all others
(b) Other deprivations in a human life have nothing to do with lack of purchasing power
(c) Incomes poverty is not a permanent condition, it changes from time to time
(d) Income poverty restricts human choices only at a point of time

Ans. (a) : According to the passage, income poverty is only one measure of counting the poor because it talks of only one kind of deprivation and ignoring all others.

14. What does the author mean by 'poverty of a life'?

(a) All deprivations in a human life which stem not only from lack of income but lack of real opportunities
(b) Impoverished state of poor people in rural and urban areas
(c) Missed opportunities in diverse personal circumstances
(d) Material as well as non-material deprivations in a human life which restrict human choices permanently

Ans. (a) : By 'Poverty of a life' the author means that all deprivations in a human life, which stem not only from lack of income but lack of real opportunities.

15. X and Y run a 3 km race along a circular course of length 300m. Their speeds are in the ratio 3:2. If they start together in, the same direction, how many times would the first one pass the other (the start-off if not counted as passing)?

(a) 2 (b) 3
(c) 4 (d) 5

Ans. (b) : Number of rounds made by each

$= \frac{3000}{300} = 10$

In the same time x takes 3 rounds, y takes 2 rounds
When they both meet, then number of rounds made by them = 3 : 2
6 : 4
9 : 8
So, hence such a situation is possible three times in total.

16. If the order of the letters in the English alphabet is reverses and each letter represents the letter represents the letter whose position is occupies, then which one of the following represents 'LUCKNOW'?

(a) OGXPMLD
(b) OGXQMLE
(c) OFXPMLE
(d) OFXPMLD

Ans. (d) : Here opposite letter–
LUCKNOW→OFXPMLD

17. In a tournament of Chess having 150 entrants, a player is eliminated whenever he loses a match. It is given that no match results in a tie/draw. How many matches are played in the entire tournament ?

(a) 151 (b) 150
(c) 149 (d) 148

Ans. (c) : Only one person won the chess tournament.
So total 149 games were played.

18. How many 3-digit natural numbers (without repetition of digits) are there such that each digit is odd and the number is divisible by 5 ?

(a) 8 (b) 12
(c) 16 (d) 24

Ans. (b) : The number is divisible by 5. So 5 will be the units place. 1, 3, 7, 9 in place of hundred is a possibility. The probability that any three out of 1, 3, 7, 9 will received in tenth place.
So, total number = 4 × 3 × 1 = 12

19. Consider the Question and two statements given below:

Question : Is x an integer?

Statement-1 :x/3 is not an integer

Statement-2 : 3x is an integer.

Which one of the following is correct in respect of the Question and the Statement

(a) Statement-1 alone is sufficient to answer the Question
(b) Statement-2 alone is sufficient of answer the Question
(c) Both Statement-1 and Statement-2 are sufficient to answer the Question
(d) Both Statement-1 and Statement-2 are not sufficient to answer the Question

Ans. (d) : 1st Statement

$\frac{x}{3}$ = Fraction

x = 3 × (Fraction)

So x is a fraction number
2nd statement